

PENGFEI HU

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Address: 1 Castle Point Terrace, Hoboken, NJ 07030

Education

Stevens Institute of Technology

Sep. 2023 – Present

Ph.D. in Computer Science (GPA: 3.95/4.00)

Hoboken, NJ

- **Ph.D. Advisor:** [Dr. Yue Ning](#);
- **Research Interests:** AI for Healthcare, AI for Environmental Science, Retrieval-Augmented Generation, Domain Generalization & Adaptation, Large Language Models, Agents, Graph Representation Learning.

University of Southern California

Sep. 2021 – May 2023

M.S. in Analytics (Industrial System Engineering, GPA: 4/4)

Los Angeles, CA

- Research Assistant advised by [Dr. Sze-chuan Suen](#) working on monitoring of diabetes via wearable devices.

Nanjing University of Finance and Economics

Sep. 2016 – May 2020

B.A. in Economic Statistics (Rank: 1 / 145, GPA: 3.63/4.00)

Nanjing, China

- Working on relative deprivation in the digital era ([one publication](#)) supervised by [Dr. Qinghai Li](#).

Work Experience

Oak Ridge National Laboratory

May 2025 – Aug. 2025

Graduate Research Intern, Mentor: Ming Fan, Leader: Dan Lu

Oak Ridge, TN

- Utilized the Frontier supercomputer server to deploy parallel training on over 10 compute nodes via DDP and mpi4py;
- **Project 1:** Develop adaptive graph encoder for multi-reservoir inflow forecasting in the Upper Colorado River Basin;
- **Project 2:** Explore feature modulation with meta attributes for enhancing model generalization across reservoirs;
- **Outcome:** One journal paper in [Environmental Modelling & Software](#) and one paper under review.

Alibaba Group, Inc.

Mar. 2021 – Jul. 2021

Data Analyst Intern

Hangzhou, China

- Tracked 20GB delivery data via Hive SQL to evaluate performance like freight volume and penetration of shipment to filter potential counties, calculated ROI and compared confident intervals to allocate ads budget efficiently by 10%;
- Summarized high-risk users and matched them with behaviour-based features (fake Mac-Address, pageview status, online duration etc.), constructed and tuned best model LightGBM by Python for prediction with 87% precision rate and saved 7% budget for fraud detection of risk control comparing with \$100k+ cost of last quarter.

Earlier Internships

- Product Analyst Intern at **Tencent** (Apr. 2020 – Aug. 2020, *Shenzhen, China*);
- Risk Data Analyst Intern at **Didi Global** (Dec. 2019 – Apr. 2020, *Chengdu, China*);
- Actuarial Analyst Intern at **PwC Consulting** (Jan. 2019 – Mar. 2019, *Shenzhen, China*).

Publications

(The google scholar link can be found [here](#), * equal contribution, † corresponding author)

1. **Pengfei Hu**, Chang Lu, Feifan Liu, and Yue Ning, “[Exploring Accurate and Transparent Domain Adaptation in Predictive Healthcare via Concept-Grounded Orthogonal Inference](#)”, Under Review, 2026.
2. **Pengfei Hu**, Xiaoxue Han, Fei Wang, and Yue Ning, “[Discovering Hierarchy-Grounded Domains with Adaptive Granularity for Clinical Domain Generalization](#)”, Under Review, 2025.
3. **Pengfei Hu**, Chang Lu, Fei Wang, and Yue Ning, “[Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning](#)”, Accepted by *IEEE Journal of Biomedical and Health Informatics (IF: 6.8)*, 2026.
4. Fan Ming*, **Pengfei Hu***, Xiaoxue Han, Wei Zhang, Hyun Kang, Yue Ning, and Dan Lu, “[AGFormer: Adaptive Spatiotemporal graph informed transformer for multi-reservoir inflow forecasting.](#)”, In *Environmental Modelling & Software (IF: 4.6)*, 2026: 106938.
5. **Pengfei Hu**, Fan Ming, Xiaoxue Han, Chang Lu, Yue Ning, and Dan Lu, “[Hydrological Domain-Conditioned Modulation for Cross-Reservoir Inflow Prediction](#)”, In *AAAI-26 AI for Environmental Science Workshop*, 2026. **Oral.**
6. Xiaoxue Han, **Pengfei Hu**, Jun-En Ding, Chang Lu, Feng Liu, Yue Ning, “[Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery](#)”, In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025.

7. **Pengfei Hu**, Fan Ming, Xiaoxue Han, Chang Lu, Wei Zhang, Hyun Kang, Yue Ning, and Dan Lu, “[Adaptive Graph Learning with Transformer for Multi-Reservoir Inflow Prediction](#)”, Accepted by *IEEE International Conference on Data Mining (ICDM) Eleventh Workshop on Data Mining in Earth System Science*, 2025.
8. Eric Yang*, **Pengfei Hu***, Xiaoxue Han, and Yue Ning, “[MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records](#)”, In *2024 IEEE International Conference on Big Data (BigData)*, 2024.
9. Hsin-Ling Hsu, Cong-Tinh Dao, ..., **Pengfei Hu**, ..., Chenwei Wu (8th/15 authors in total), “[MEDPLAN: A Two-Stage RAG-Based System for Personalized Medical Plan Generation](#)”, In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, 2025.
10. Qinghai Li, Hao Li, and **Pengfei Hu**[†], “[The impact on Chinese residents’ subjective relative deprivation in the digital era](#)”, In *Information Technology for Development (IT: 6.4)*, 2025: 1172-1194.

Technical Skills

- **Programming Languages:** Python, SQL, Java, R, Latex.
- **Deep Learning Framework:** Pytorch, Keras.

Honors and Awards

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| <ul style="list-style-type: none"> • Stevens Institute of Technology Provost Doctoral Fellowship • University Outstanding Graduate (Summa Cum Laude) • Highest Honor in Honor Thesis • National Scholarship of Zhongli International (Top 0.5%) • Dean’s List | <p><i>Fall 2023, Spring 2024</i></p> <p><i>Spring 2020</i></p> <p><i>Spring 2020</i></p> <p><i>Fall 2019</i></p> <p><i>Fall 2018, Spring 2019, Fall 2019, Spring 2020</i></p> |
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Services

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| <ul style="list-style-type: none"> • Teaching Experience: <ul style="list-style-type: none"> • Teaching Assistant for CS-584: Natural Language Processing. • Teaching Assistant for ISE-535: Data Mining. • Research Experience: <ul style="list-style-type: none"> • Research Assistant in USC Viterbi School of Engineering. • Research Assistant in Center for Economic and Social Research (USC Dornsife). • Graduate Researcher in Oak Ridge National Laboratory (GRO). • Authorized User of the Frontier Supercomputer at the Oak Ridge Leadership Computing Facility. • Conference and Journal Review: ICML 2026; ACL Rolling Reviews (bimonthly since July 2025); KDD 2026; AAAI 2026; MICA I 2025, Computers in Biology and Medicine, IEEE Access. | <p><i>Fall 2024, Spring 2025</i></p> <p><i>Spring 2022, Fall 2022</i></p> <p><i>Summer 2022</i></p> <p><i>Fall 2022, Spring 2023</i></p> <p><i>Summer 2025</i></p> |
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